Presentation Script — Telecom Data Analytics Capstone

Slide 1 — Introduction

Good morning everyone.

Today, we are presenting our final capstone project titled **“Telecom Data Analytics: From Raw Data to Business Recommendation.”**

Our project is an end-to-end AI and ML based telecom analytics system. The main idea is to take raw and messy telecom data, clean and process it, apply machine learning and analytics, and finally convert the results into useful business decisions.

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Slide 2 — Executive Summary

The main goal of this project is to convert messy telecom operational data into **reliable insights and actionable recommendations**.

We combined several areas such as data engineering, machine learning, forecasting, NLP, monitoring, and business analysis.

The process starts with raw data ingestion and profiling. Then we clean and validate the data to create a governed dataset.

After that, we apply different analytical and machine learning techniques. Finally, the results are converted into dashboards, alerts, and cost-based recommendations.

So, the key point is that this project is **not just an ML model**. It is a complete decision system that connects **data to insight, insight to action, and action to measurable business value.**

Slide 3 — What Problem Are We Solving?

A telecom company generates a huge amount of data from subscribers, network systems, usage, revenue, and customer support.

But this raw data is not always reliable. It can contain duplicates, missing values, inconsistent information, and operational noise.

Because of this, simply applying machine learning to the raw data may produce misleading results.

We focused on four major business questions.

First, **subscriber risk** — which customers are likely to churn and which customers are worth retaining?

Second, **network quality** — which network cells are degrading, and how much revenue or customer value is affected?

Third, **customer care** — which customer messages are negative and which support tickets should be routed automatically?

And finally, **business value** — where should the company invest and which action can create the highest value?

So our overall objective is to move from **data → insight → action → measurable value.**

Slide 4 — Telecom Data Used

For this project, we worked with multiple telecom datasets.

The subscriber dataset contains customer profiles, six months of behavior, and churn outcomes.

The usage dataset contains monthly usage and revenue information.

For network analysis, we used network KPI data, maintenance windows, and monthly cell traffic and capacity data.

For customer care, we used customer messages with sentiment labels and support ticket information.

Each dataset has a different level of detail. For example, subscriber data is at the subscriber level, network KPI data is at the cell and 15-minute level, and messages and tickets are recorded individually.

This difference in data grain is important because it affects how we clean, join, and analyze the data.

Slide 5 — Why Data Governance Comes Before Modelling

Before building any machine learning model, we first need to make sure that the data is trustworthy.

We found several data quality problems.

For subscribers, there were duplicates, missing usage and recharge values, and inconsistent region names.

The usage data contained duplicate records and missing or inconsistent revenue values.

The network data had duplicate readings, blank latency values, negative throughput, maintenance outages, and collector losses.

There were also customer messages that did not have matching subscriber records.

Instead of simply deleting all bad records, our approach was to first understand **what the problem actually means** and then decide how to repair it.

That is the main principle of our data governance process: **do not blindly delete bad data; understand it before repairing it.**

Slide 6 — From Raw Data to a Governed Dataset

We converted the raw CSV files into a more reliable and governed dataset.

The main steps were:

First, identify the correct data grain.

Then detect duplicates and remove them at the correct level.

After that, classify missing values before deciding how to handle them.

We also standardized categories such as region names.

When we imputed missing values, we added a **was_missing flag**, so we could still track which values were originally missing.

Some unusual values needed special treatment. For example, negative throughput was treated as a **counter rollover** instead of simply deleting those records.

Similarly, NULL latency during planned maintenance was preserved because it represented a real operational condition.

Finally, the pipeline was made **idempotent**, meaning that if we run it twice, we get consistent results instead of creating duplicate data.

Slide 7 — Preventing Data Leakage

Another important part of our machine learning process was preventing **data leakage**.

Data leakage happens when a model gets information that would not actually be available at the time we make the prediction.

We used what we called the **Day-Before Test**.

For every feature, we asked:

“Would this value have existed the day before the event we are trying to predict?”

If the answer was no, we considered it leakage.

For example, we removed the feature **retention_offer_sent**, because a retention offer could already be a result of identifying a customer as high risk.

We also removed **total_charges**, because it can effectively reveal information about ARPU and tenure and therefore indirectly reveal the target.

The important lesson is that a high model accuracy is not useful if the model is using information that would not be available in the real world.

Slide 8 — Three Analytical Tracks

We divided the project into three main analytical tracks.

Track A is Churn and Retention.

Here, our question is: which subscribers are likely to leave, and how much should we spend to retain them?

We used behavioral features, churn classification, segmentation, expected loss, and anomaly detection.

Track B is Network and Revenue.

Here, we ask which network cells are degrading, what revenue is exposed, and where additional capacity should be added.

This includes KPI cleaning, alert engineering, forecasting, revenue exposure, and capacity planning.

Track C is the Customer Care Copilot.

The goal here is to understand which customer messages are important, route tickets to the appropriate team, and prioritize customers based on value.

For this track, we use sentiment analysis, ticket routing, confidence thresholds, human escalation, and customer-value prioritization.

Slide 9 — Track A: Churn Prediction and Segmentation

In Track A, we identified a **priority retention segment**.

This segment contains **392 subscribers**, which represents about **12.2% of the customer base**.

The churn rate for this group is around **40%**, compared with **24.1% overall**.

Interestingly, the segment is mainly driven by customer complaints rather than simply high spending.

We also calculated expected loss using:

Expected Loss = Churn Risk × ARPU × 12

This helps us prioritize customers not just based on prediction accuracy, but based on their actual business value.

We also used Isolation Forest for anomaly detection to identify possible fraud, subscription issues, or billing faults.

However, an anomaly flag is not treated as an automatic conclusion. It simply tells us that the case requires further investigation.

Slide 10 — Track B: Network Quality and Revenue Optimization

Finally, in Track B, we focused on network quality and revenue impact.

After cleaning the network data, we reduced **8,681 raw rows to 8,640 rows** after removing duplicates.

We also analyzed the blank latency values. Out of 97 blank values, some were caused by planned maintenance, some by outages, and others by collector loss.

We identified **9 negative throughput values**, which were treated as counter rollovers, and **27 downtime rows**.

We also identified three cells operating above **85% utilization**:

CELL-HYD-03, CELL-KOL-03, and CELL-MUM-01.

The key business lesson here is that network problems should not be prioritized only based on technical metrics like latency or utilization.

Instead, we should also consider **customer impact and revenue exposure**.

So, the final goal of our system is not just to find problems, but to identify **which problems matter most to the business and where action should be taken first.**

Closing

To conclude, our project demonstrates a complete journey from **raw telecom data to business recommendations**.

We first improved data quality and governance, then applied machine learning and analytics to three major business areas: **customer churn, network and revenue optimization, and customer care**.

The main takeaway is that AI and ML become much more useful when they are connected with reliable data, proper business logic, monitoring, and measurable actions.

Thank you. We would be happy to answer any questions.